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ORIGINAL ARTICLE

# Improving Primary School Students' Oral Reading Fluency Through Voice Chatbot-Based AI

Mohamed Ali Nagy Elmaadaway[1,2] | Mohamed Elsayed El-Naggar[3] | Mohamed Radwan Ibrahim Abouhashesh[2]

[1]Deanship of Scientific Research, King Saud University, Riyadh, Saudi Arabia | [2]Faculty of Specific Education, Department of Educational Technology and Computer, Kafr El-Sheikh University, Kafr El-Sheikh City, Egypt | [3]Faculty of Educational Studies Programs Director, National Egyptian E-Learning University (EELU), Cairo City, Egypt

**Correspondence:** Mohamed Ali Nagy Elmaadaway (melmaadaway@ksu.edu.sa; prof_nagy@spe.kfs.edu.eg)

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## ABSTRACT

**Background:** Artificial intelligence (AI) made substantial progress with language recognition. Proficiency in spoken English reading is a prerequisite for fluency in written English. However, research on its use, especially for non-native speakers, is lacking despite increased usage.

**Objectives:** This study aimed to enhance the oral reading fluency abilities of fourth graders using the AI voice chatbot 'Alexa', the purpose being to examine the effects of Alexa on fourth graders' oral reading fluency skills.

**Methods:** Ninety students were evenly divided into experimental and control groups. We developed, reviewed and administered pre- and post-tests and interviews.

**Results and Conclusion:** Students' reading comprehension and oral reading fluency are enhanced when they use Alexa in the classroom. Many students turn to Alexa to help them with reading and English grammar. This study offers suggestions for future research.

## 1 | Introduction

English reading and writing are essential conversational and comprehensive skills (Dewi et al. 2021). Fluent readers read faster and recognize words naturally (Schwanenflugel and Kuhn 2016), understanding the whole rather than individual words (Rupley et al. 2020). Direct instruction and constructive criticism can boost reading fluency (McKenna and Bettini 2018).

As curricula become more tailored, artificial intelligence (AI) and machine learning (ML) algorithms in educational contexts are increasingly used (Munir et al. 2022). Research suggests that integrating AI with teaching fundamental skills and language proficiency is the most effective way to educate language users (Lehlou and Brigui 2021). However, using AI requires addressing ethics and privacy concerns, and interdisciplinary collaborations for advancement (Zhang and Aslan 2021). Srinivasan and Murthy (2021) investigated the effects of AI technology on English reading and comprehension skills. They evaluated students' proficiency levels and facilitated their learning and reading abilities through intelligent learning tools and adaptive smart content. Madnani et al. (2019) demonstrated benefits of AI in enhancing reading fluency and comprehension through a digital application, 'My Turn to Read', which alternates between reading independently and listening to text read aloud and can record the readings.

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**Summary**

- What is already known about this topic
  - Primary school students in Egypt struggle with learning a second foreign language (English)
  - Artificial intelligence (AI) applications, especially voice chatbots, are of great importance in developing language skills for students.
  - Amazon Alexa is considered one of the effective and easy-to-program applications that can contribute to the development of English language skills.
- What this paper adds
  - It provides a deeper understanding of the practical application of AI tools like Amazon Alexa for improving primary school pupils' oral reading fluency.
  - It enhances students' learning with AI applications by exploring the impact of using voice chatbot-based AI (Alexa) for research purposes.
- Implications for practice and/or policy
  - The study underscores the potential of incorporating Amazon Alexa, an AI chatbot, in English language instruction to address weaknesses in oral reading fluency and reading comprehension among elementary school students in Egypt.
  - Findings demonstrate that the integration of AI technology holds promise for enhancing language proficiency and addressing academic shortcomings.
  - Educators and instructors can utilize the study's outcomes and criteria for Alexa to develop, implement and assess technological interventions in English language education.
  - Proficiency in using AI technology is crucial for educators to create instructional materials effectively.
  - The research methodologies employed in this study can serve as a model for developing similar software applications, providing inspiration and guidance for instructors.
  - Training sessions for English language educators can be conducted to gauge their perspectives on integrating Alexa into instructional practices, with subsequent publication of findings.
  - Future research could explore the utilization of chatbots across different educational levels and disciplines, including secondary and higher education, for enhancing research capabilities and fostering higher-order cognitive abilities.

AI chatbots are effective tools for language instruction—students can improve language skills at their convenience and receive personalized feedback in convenient learning environments (Guo et al. 2023; Jeon et al. 2023). Chatbots can improve students' motivation, engagement and academic results (Fathi et al. 2024; Kim and Su 2024; Ou et al. 2024). AI voice chatbots simulate human conversation, offering an interactive and engaging platform for learning (Jeon et al. 2023). Society is adopting voice chatbots such as Google Assistant and Amazon's Alexa (Terzopoulos and Satratzemi 2020; Wu et al. 2024), Alexa being the most popular (Belda-Medina and Calvo-Ferrer 2022). According to Tai and Chen (2024), voice chatbots can offer diverse topic discussions tailored to learners' interests, making them effective for elementary school students with limited English proficiency. This adaptability is crucial in maintaining student engagement and motivation as learners are more likely to invest in activities they find relevant and interesting. Voice-based chatbots in English language learning show significant potential, with recent studies focusing on theoretical frameworks, methodological and technological properties, user reports and pedagogical implementations (Koç and Savaş 2024).

The advantages of speech chatbots to learn English lie in their input and output accessibility. Ali (2020) examined the efficacy of an AI application to enhance oral language abilities, specifically listening comprehension and speaking skills, among sixth-grade primary school students. The researcher employed chatbots as a means of developing speaking and listening abilities while overseeing and analysing their interactions, and taught educators on the optimal utilization of chatbots as a pedagogical tool to enhance students' learning outcomes.

The effectiveness of AI voice chatbots in improving reading fluency is the primary focus of this study. Yuan (2023) illustrated that AI chatbots can enhance spoken language skills, which can enhance reading fluency. By engaging in conversations with chatbots, students practise pronunciation, intonation and pacing, all of which contribute to improved oral reading skills. Han (2020) highlighted the positive impact of chatbots on learners' attitudes towards English learning, suggesting that they can foster a positive learning environment conducive to skill development.

This study compares the outcomes of students in the experimental group, using AI voice chatbots, with those in the control group, following traditional reading practices. Liu et al. (2022) highlighted the variability in outcomes depending on the methodologies employed and the characteristics of the sample population. By maintaining a standardized approach across both groups, this study provides consistent and reliable data on the effectiveness of AI interventions. Understanding students' qualitative experiences with AI chatbots is crucial to evaluating their effectiveness. Interviews with students provided insights into their perceptions of AI, including ease of use, relevance of the content and nature of feedback received. Hoang et al. (2023) noted students' willingness to engage with AI tools is often influenced by their perceived usefulness and ease of use. The study captured a diverse range of experiences, recognizing that students have different learning styles and preferences. By exploring these experiences, this study identified best practices for implementing AI chatbots in educational settings and addressing potential barriers to their effective use. There are challenges and limitations in using voice chatbots. Technical issues, such as voice recognition errors and software glitches, can hinder the learning experience and frustrate users. Azizimajd (2023) accentuated the importance of addressing these technical challenges to ensure smooth and effective interaction between students and chatbots.

To fully understand the context and significance of this study, one should recognize that in many non-English-speaking countries, including Egypt, students are required to learn EFL. Many of these students struggle with key language skills. Frequent oral reading is one of the most effective methods for improving comprehension, fluency and accuracy (NICHD 2000).



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This study examined the potential of AI-powered voice chatbots as a transformative tool for enhancing oral reading fluency among primary school students, seeking to contribute to the literature on the integration of AI in education. The findings provide valuable insights for educators and policymakers, demonstrating how AI tools can be leveraged to improve reading instruction and overcome persistent challenges in language learning.

This study's implications are far-reaching, as they inform the design, implementation and refinement of AI technologies in classroom settings. By highlighting the opportunities and challenges of using AI in education, this research underscores the potential of AI to create more engaging, personalized and effective learning experiences for students. Given the limited research on AI speech systems and their impact on second-language oral reading skills, this study fills critical gaps in the existing knowledge, specifically the influence of chatbot systems on the accuracy and fluency of elementary school students' oral reading in a second language, offering practical and evidence-based solutions for improving language education.

## 2 | Literature Review

### 2.1 | Oral Reading Fluency

Oral reading fluency considers the number of errors readers make when reading a passage aloud. This includes the tracking of incorrect and correct words, repeated words or phrases and speed (Aldhanhani and Abu-Ayyash 2020). Reading fluency is the ability to read texts effortlessly, accurately and to identify and remedy errors in subsequent readings (NICHD 2000). The process of word recognition commences with decoding; readers need strong automatic decoding skills to comprehend the text's ideas rather than individual letters and words (Armbruster et al. 2003; Hudson et al. 2005). Fluency, a higher-order skill (Hasbrouck and Tindal 2006), includes decoding, word-meaning detection, phrase structure, syntactic processing and inference drawing.

Reading is a skill required for cognitive and educational development (Chall 1996). Fundamental skills (e.g., phonemic awareness, familiarity with letter sounds and decoding abilities) (Melby-Lervag et al. 2012) serve as the foundation for advanced skills such as fluency and comprehension. Phonological awareness and phonics—crucial for effective reading instruction—involve the development of skills related to the recognition and understanding of connections between sounds and symbols, and the acquisition of vocabulary, fluency and comprehension abilities (Armbruster et al. 2003; Linan-Thompson and Vaughn 2004).

Comprehension of spoken language requires more than merely identifying words—establishing a connection between each word and its corresponding semantic representation is needed to grasp the intended meaning of the text (Nation 2005; Kamhi and Catts 1991; Rayner et al. 2001). Vocabulary proficiency encompasses the capacity to employ words in several communicative domains, including expressive modes (e.g., speaking, writing) and receptive modes (e.g., listening, reading). Nagy and Scott (2000) opined that comprehensive understanding can only be achieved once approximately 90%–95% of its vocabulary has been committed to memory. The impact and predictive nature of students' vocabulary on their future reading comprehension are well documented (Muter et al. 2004; Roth et al. 2002; Share and Leikin 2004).

Research established a connection between reading comprehension and fluency, particularly among novice readers and individuals acquiring literacy skills in a familiar language (Fuchs et al. 2001). Fuchs et al. (2001) found a significant positive correlation (0.91) between the Stanford Achievement Test's reading comprehension and assessments of oral reading fluency, determined through timed evaluations of the number of correct words read per minute. Including oral reading fluency in comprehension subtasks plays a critical role in facilitating understanding (Martins and Capellini 2019). According to Kuhn et al. (2010), fluency encompasses components such as emotions, precision and quickness, specifically in terms of prosody. The oral reading–fluency subtask is a timed assessment that quantifies reading speed and accuracy by measuring the number of correct words read per minute, but it does not evaluate the use of expressions (Piper and Zuilkowski 2015).

### 2.2 | AI and Learning Languages

Computer-assisted language learning (CALL) is well researched, but the usage of mobile applications, online platforms and AI-driven solutions has increased considerably. Students previously utilized computer-based platforms such as radio applications, YouTube and podcasts to study foreign languages (Karkera and Chamundeshawari 2018; Yoestara and Putri 2019). The integration of AI features, such as self-correction and human interaction, augments the significance of humanoid media.

AI can boost individuals' motivation and offer supplementary avenues for engagement and communication, thereby facilitating oral proficiency. According to Becker and Edalatishams (2019), the integration of AI technologies such as ELSA Speak into voice recognition systems can improve verbal communication and decrease word errors. Zou et al.'s (2020) survey determined participants' views on the effectiveness of mobile AI-assisted applications in improving their English speech—the data indicate students' predilection for AI to improve their speech. AI is a motivator because of its feedback, encouraging them to actively participate. According to Kim (2018), the use of AI in educational environments can augment students' oral fluency.

AI facilitates English acquisition via tools such as Google Assistant, Apple's Siri, Samsung's Bixby and Amazon's Alexa (Cambre et al. 2020). Virtual reading aids can enhance reading fluency; Madnani et al. (2019) developed an e-book reading tool, 'My Turn to Read', allowing students to listen to text read aloud. Its replay function assists students in concentrating on a section and using the voice-recording technology to assess their pronunciation, facilitated by a virtual reading companion that enhances reading fluency and comprehension abilities.

Bryant et al. (2015) employed an alternating-treatment design to examine differences in academic engagement and outcomes,



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specifically word identification and reading fluency, between AI and traditional teacher-directed instruction (TDI). Both instructional approaches were effective in promoting student engagement, but students exhibited a higher level of engagement with the AI method, whereas visual and quantitative analyses indicated that TDI exhibited greater efficacy than AI regarding passage fluency and word recognition.

### 2.3 | The Impact of Chatbot Programs on Second-Language Skills

Computer programs that can understand and respond to voice instructions and inputs provided in natural language are known as voice chatbots, voice assistants or virtual assistants (Athikkal and Jenq 2022). Technological advancements enable chatbots to improve both functionality and user experience, making them an essential part of the technology industry (Gomathy et al. 2021).

First to examine how intelligent personal assistants (IPAs) improve hearing and speaking was Dizon (2020), who adopted Amazon Echo Dot. The research sample comprised 13 experimental and 15 control participants. Ten weeks of 12-min sessions were given to the experimental group. Using Amazon Echo Dot had a beneficial impact on students' speaking proficiency. No disparity emerged on the enhancement of listening abilities between the groups that used IPAs and those that did not. Despite Echo Dot's IPAs' positive outcomes, generalization of the results was constrained by the small sample size, the brief duration of the experiment and the absence of visual aids, which significantly influence learning. Yang et al. (2022) involved more participants to yield more dependable results. They developed 'Ellie', a task-based speech chatbot. Researchers tested Ellie as an English conversation companion and examined students' opinions on using it in an EFL lesson. Ellie assisted 314 Korean EFL learners aged 10–15 who completed three speaking challenges in class. With the original 1000-word band, the participants averaged 9.63 revolves per session, demonstrating that the chatbot encouraged student participation—a rarity in Korean EFL classes. Although Yang et al. (2022) used a large sample, the age gap among participants may affect their prior computer program and linguistic skills, reducing the reliability and generalizability of the findings.

To improve listening and speaking ability, Chien et al. (2022) developed a novel contextual English learning environment, utilizing the communication program LINE Chatbot, based on the AI Mark-up Language (AIML). Seventy-three students participated in a 4-week English discussion exercise that included listening and speaking. This contextual learning environment promoted their English speaking and listening. Comparable outcomes were achieved with identical methodology by Wu et al. (2024), who proved that IPAs significantly enhanced second-language skills by using an IPA, 'MSLIPA'. The findings demonstrate that the experimental group exhibited improved conversational skills and listening and speaking competency in the second language relative to the control group.

Ignorance of Chatbot-based language improvement technology can cause anxiety among learners. Wu et al. (2024) suggested that future studies should familiarize learners with IPA technology before interventions to ensure they perceive it as a collaborating tool. Regarding fear, anxiety and proficiency in communicating, Kim and Su's (2024) investigation used a chatbot application that could enhance proficiency in communicating in Korean as a secondary foreign language. Their findings revealed that all in the experimental group had reduced levels of fear and increased confidence in their communication skills. The findings demonstrate that interactions facilitated by chatbots created a learning environment favourable for student learning, with heightened levels of engagement and enhanced learning experience. Alrajhi (2024) employed a mixed-methods approach to investigate EFL students' perceptions of a chatbot that utilized text-based instruction. Their findings suggest that the chatbot's intelligibility and comprehension were positive. The chatbot facilitates L2 practice and enhances writing skills while motivating users, thereby alleviating writing anxiety. Constraints to the chatbot's effective communication and education included brief conversations, sensitivity to inappropriate language and irrelevant responses. Kwon et al. (2023) investigated how chatbot-based writing practices affected the writing attitudes of 75 Korean elementary school pupils as second-language learners. The experimental group practiced writing with a chatbot for 15 weeks, whereas the control group received teacher-led instruction. Google's Dialogflow machine-learning AI tool encoded elementary school English textbook utterances to create the chatbot. Pretest scores revealed similar writing proficiency, but the experimental group performed much better on the post-test than the control group, and they found the chatbot enjoyable.

Regarding the potential of voice chatbot-based AI in SLA, Yuan (2023) found that chatbots with distinct voices and tailored interactions significantly improved primary school students' spoken English. The study highlighted the importance of incorporating diverse topic discussions and personalized feedback, which accommodated learners' interests and proficiency levels. Such customization increased student engagement.

AI's use in improving reading fluency is supported by empirical evidence and grounded in theories that emphasise the importance of personalised learning. Theories of constructivism and cognitive load suggest that students learn best when they actively engage with material that is neither too easy nor too difficult (Sweller 1988). AI technologies can dynamically adjust the complexity of reading materials, aligning with these theories to optimize learning outcomes (Luckin et al. 2016). This study anticipated students in the experimental group would exhibit greater improvement in reading fluency than those in the control group. This hypothesis was based on previous studies, such as Koç and Savaş (2024), who demonstrated significant gains in language proficiency among students using AI chatbots compared to those using conventional methods. The analysis focused on metrics such as reading speed, accuracy and expression, providing a comprehensive assessment of oral reading fluency.

The utilization of AI voice chatbots in SLA is grounded in several theoretical frameworks, including the Input Hypothesis and Interactionist Theory. Krashen's (1985) Input Hypothesis posits that language acquisition occurs when learners are exposed to



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comprehensible input slightly above their current proficiency level (i + 1). Voice chatbots can deliver such input by tailoring conversations to individual learners' abilities, promoting gradual language development (Tai and Chen 2024).

Interactionist Theory emphasizes the importance of interaction in language learning (Long 1996). Through real time conversations, chatbots allow learners to practice speaking and listening skills in a non-threatening environment (Ali 2020). This aligns with negotiated interaction—learners modify their output based on feedback, leading to enhanced language proficiency (Mackey 2007). Studies indicate that interactive digital environments can replicate aspects of face-to-face communication, facilitating language practice and development (Satar and Ozdener 2008).

Research on chatbots in language acquisition demonstrated their potential and effectiveness, but no study to date has specifically evaluated oral speech across its four critical dimensions: fluency, accuracy, vocabulary and comprehension, and also overall oral reading fluency skills. While IPAs and chatbots have been employed to improve second-language skills, their impact on oral reading accuracy remains underexplored, particularly among Arabic-speaking students. This study bridges this gap by targeting Arabic-speaking learners' English oral reading accuracy. The research involves a homogeneous sample of primary school students with shared cultural and linguistic backgrounds without prior experience learning EFL.

Previous studies often lack detailed explanations of the methodologies used to develop chatbot programs aimed at enhancing language abilities. This study consequently provides a detailed account of the procedure involved in designing and implementing a chatbot program using Alexa, which offers valuable guidance for researchers and educators interested in leveraging chatbot technology for language instruction. It addresses the existing gap in the literature by investigating the effects of AI chatbots on oral reading fluency among elementary school students in Egypt through a rigorous experimental design and explores students' perceptions of using chatbot applications for learning EFL. The study was guided by the following research questions:

RQ1. *To what extent does the use of AI chatbots improve primary students' oral reading fluency?*

RQ2. *What are the participants' views on the use of AI-powered chatbot applications for learning EFL?*

## 3 | Methodology

### 3.1 | Participants

The study sample comprised 90 fourth grade elementary school children. Participants were of mixed genders, aged 9–10 years (Table 1). They had just begun learning EFL, which was the reason for selecting this sample. All students were Egyptians, with Arabic as their native language.

**TABLE 1** | Sample details.

| Group | Gender | Ratio | Age average |
|---|---|---|---|
| Experimental | Male | 44.44% | 9.95 |
| | Female | 55.56% | |
| Control | Male | 48.89% | 9.86 |
| | Female | 51.11% | |

Teachers encounter challenges in teaching EFL, particularly in oral communication, as evidenced by the observations and oral test results. Consequently, students' oral skills were augmented by AI-based chatbot applications.

Participants were divided into two groups of 45 students each. The experimental group used an AI chatbot to study English, whereas the control group followed traditional methods.

Table 1 shows a convergence in the proportions of boys and girls and age range in both the experimental and control groups.

### 3.2 | Instruments

The instruments included chatbots developed using the Alexa AI platform, an oral reading fluency test and semi-structured interviews.

#### 3.2.1 | Oral Reading Fluency Test

The oral reading fluency test is used to evaluate students' reading skills. The achievement examination for reading comprehension and oral reading was based on Katz and Carlisle (2009). It comprised a sequence of main and secondary tasks designed to assess fundamental proficiency in spoken reading. The assessment tool is methodologically impartial, without bias regarding literacy training approaches. To achieve reading proficiency, expertise in phonological awareness, phonics, vocabulary, fluency and comprehension must be acquired (Armbruster et al. 2003; Linan-Thompson and Vaughn 2004).

To ensure the validity and reliability of this test, it was administered by five professors with expertise in English language instruction, assessment strategies and second-language acquisition. Their involvement guaranteed that the assessment tool was linguistically precise, methodologically robust and adaptable to various learning styles and linguistic competencies. The test design was refined per their feedback, ensuring alignment with established theoretical frameworks and practical considerations in oral reading fluency assessment (Fuchs et al. 2001; Rasinski and Samuels 2011; Kuhn and Schwanenflugel 2008).

The internal consistency of the test was determined by calculating the correlation coefficient between each item and the overall score. The correlation coefficients observed in this study ranged from 0.546 to 0.871, suggesting an adequate level of internal



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consistency; thus, it was considered valid. By computing the correlation coefficient between each item and the overall score, we successfully determined the degree of compatibility among the items.

To assess the reliability of the oral reading fluency test, researchers administered it to a sample of 19 students. The ease coefficient was 0.648, reflecting the level of ease experienced by participants. The difficulty coefficient of the test for students was 0.35. The reliability of the test was evaluated using Cronbach's alpha, yielding a coefficient of 0.954. Therefore, the test and its constituent elements have a high degree of reliability and can be effectively employed to examine the sample.

The oral reading fluency test was administered in a single student–examiner setting. Tests in the subsequent sequence were performed as a constituent of the passage, delineating an evaluation that assessed the student's aptitude in comprehending, fluent reading and accurately interpreting text corresponding to their grade level. Students had to read a narrative in 1 min and answer questions on vocabulary acquisition and reading comprehension. The assessor employed standardized grading criteria to measure the accuracy and fluency of students' reading. The test was appraised using rubric descriptors, encompassing correctness, automaticity, speed, prosody, expression and pronunciation. The highest attainable score was 16.

Vocabulary comprehension was tested after the reading. The reading material was then extracted, and students were probed about its newly introduced words. Vocabulary assignment was assessed by assigning scores according to the accuracy of responses, the highest attainable score being four. The multiple-choice question reading comprehension section followed, testing students' ability to retain information and make inferences based on the text.

### 3.2.2 | Semi-Structured Interviews

The oral reading accuracy test results were systematically analysed to develop targeted interview questions. It involved identifying patterns in reading errors, such as substitutions, omissions, insertions or mispronunciations, providing insights into participants' decoding skills, fluency and comprehension (Keenan et al. 2008). By examining these error types, specific areas of participants' difficulties were pinpointed, and interview questions were formulated to address underlying cognitive and linguistic processes (Snowling and Hulme 2005).

Semi-structured interviews with the experimental group provided insights into voice chatbots impact on oral reading. This qualitative method echoes Vygotsky's sociocultural theory (Vygotsky 1978), which emphasizes the importance of interaction and tools in language learning. Vygotsky's theory (Vygotsky 1978) accentuates chatbots' ability to facilitate language acquisition through interaction and scaffolding. Constructivist learning theories (Piaget 1970) support the notion that students actively construct knowledge through meaningful and engaging activities (e.g., interacting with chatbots). Chatbots are an interactive medium, allowing oral reading practice in a convenient environment. The semi-structured design facilitated flexibility in questioning—useful for exploring lived experiences (Cohen et al. 2018).

Semi-structured interviews are used to explore students' perceptions of technology-enhanced learning environments (Creswell and Poth 2018; Miles et al. 2014). They provide rich, contextual data, offering nuanced insights into learners' experiences and attitudes. By combining deductive analysis with thematic categorization, this study ensures methodological rigour and credible and transferable findings.

Participants were asked two questions:

1. What is your perspective on utilizing a chatbot to augment your reading proficiency?
2. Has engaging in AI voice chatbot activities enhanced your reading proficiency? Why and how?

To create a comfortable environment and encourage in-depth reflections, interviews were conducted in Arabic, lasting 15–25 min. The native language ensured accuracy, reduced anxiety and fostered unrestricted responses (Birks and Mills 2015).

### 3.2.3 | Voice Chatbot-Based AI Alexa Application Design

Amazon's Alexa is the AI voice chatbot application that will be the focus of future research. Alexa allows users to have unrestricted conversations, enabling language students to practice their English conversational skills by conversing with Alexa's social bot.

The advantage of combining reading aloud with Alexa is that it can be used as a reading partner. Further to Alexa pronouncing words and sentences, students may receive feedback concerning the errors they make while reading aloud. Students can select the role to read (Figure 1).

A working prototype of the specialized Alexa skill was produced using the Amazon Developer Console. Amazon Alexa's app on a mobile device tested the developed skill. Subsequently, the logic of the application and voice interface required to connect with Alexa was built and placed in a user-defined interaction model (Figure 2).

Users' voice input was assigned to the 'intent' to construct the speech interface, enabling Amazon cloud-based services to process voice input. Students were tested and graded using the Alexa Skill Prototype. The skills were developed using the AWS Lambda function to host the developed skill. AWS Lambda is a service that runs codes on the cloud without the need to manage servers. This is performed using Node.js (Appendices A and B in Supporting Information).

Application development included application design, interface and database; creating the programming language; and locating the data storage. The application was designed as follows:



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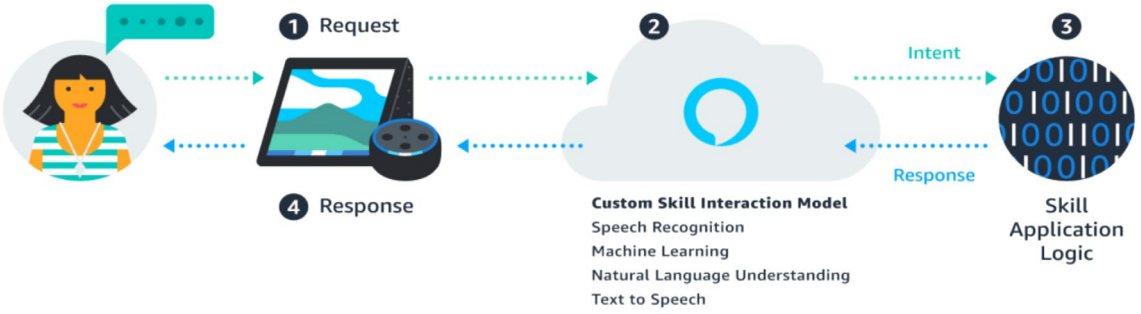



**FIGURE 1** | Alexa voice-activated processing flow.




**FIGURE 2** | Skill development workflow.

1. Types of media (reading text, words and sentences).
2. The nature of the assignment participants had to complete determined the activity level and types that would be generated.
3. Information and abilities acquired as students completed oral reading fluency exercises.
4. Feedback methods used to assess participants' comprehension. Alexa was tested on a pilot sample before applying them to the group.

Once skill coding was completed, 'English Fluency Reading Skills' was tested and ensured it could be submitted to obtain an Amazon Certificate (Appendices C, D and E in Supporting Information).

Testing began with a short introduction to oral reading fluency and Alexa Skill, and students were asked to read with Alexa.

The Alexa history page saved spoken commands as audio files and gathered information about students' usage of voice chatbot-based AI. This information was compiled from participants' commands over 12 weeks. The number of commands students provided was captured in Excel. Each command audio file was listened to and cross-checked with the stored text recorded by Alexa's voice recognition system once the total number of commands was determined. Commands with matched text and audio were classified as understood. If the text did not match the audio, it was classified as 'communication breakdown'. 'Repeat' was used when participants responded to a communication failure by repeating the same instructions.

Figure 3 shows students' usage and intent confidence in the application and their responses to the interactions with Alexa during specific periods, in addition to the daily participation rate and the rates of interaction with words and sentences at different levels.

Designers and students provided input. When teachers and students actively participate in the implementation process, quick changes can be made, increasing effectiveness and success (Manichander 2016).

Examples:

1. Students' emotional responses to the initial research demonstration (receptive or resistant).
2. How would potential testing errors be handled?
3. Was there a back-up plan if the application failed?
4. Can the student group operate autonomously once they have obtained the material, or will they need regular supervision?

The experiment was conducted in one educational area, assisted by instructors and experts, the purpose being to assess whether the objectives were achieved and to offer the resources needed to maximize success.

### 3.3 | Procedure

Forty-five fourth grade primary school students were randomly selected for the experimental group, subject to possessing smart devices with SIM cards and active phone numbers. Another 45 from the same grade were randomly selected as the control group. Experimental group children's parents agreed to their participation and using the smartphone application. Participants underwent preadministration of an English language test and, under the guidance of an educational technology professional and observation of researchers, the programme was installed on their phones and its use explained. Students selected 'English fluency reading skills' in the programme, and Alexa read words participants had to repeat. Alexa gauged students' pronunciation, correcting if necessary. Students' interactions were reviewed via Alexa's summary. Researchers and teachers regularly met with participants to ensure students benefited from the application.

Participants were evaluated on their oral reading fluency by reading one passage before and after the testing phases to track progress in speed, precision, and understanding. Students took turns reading aloud, and the examiner noted their reading comprehension, vocabulary, and any mistakes. When finished reading or having made too many reading errors, the test was



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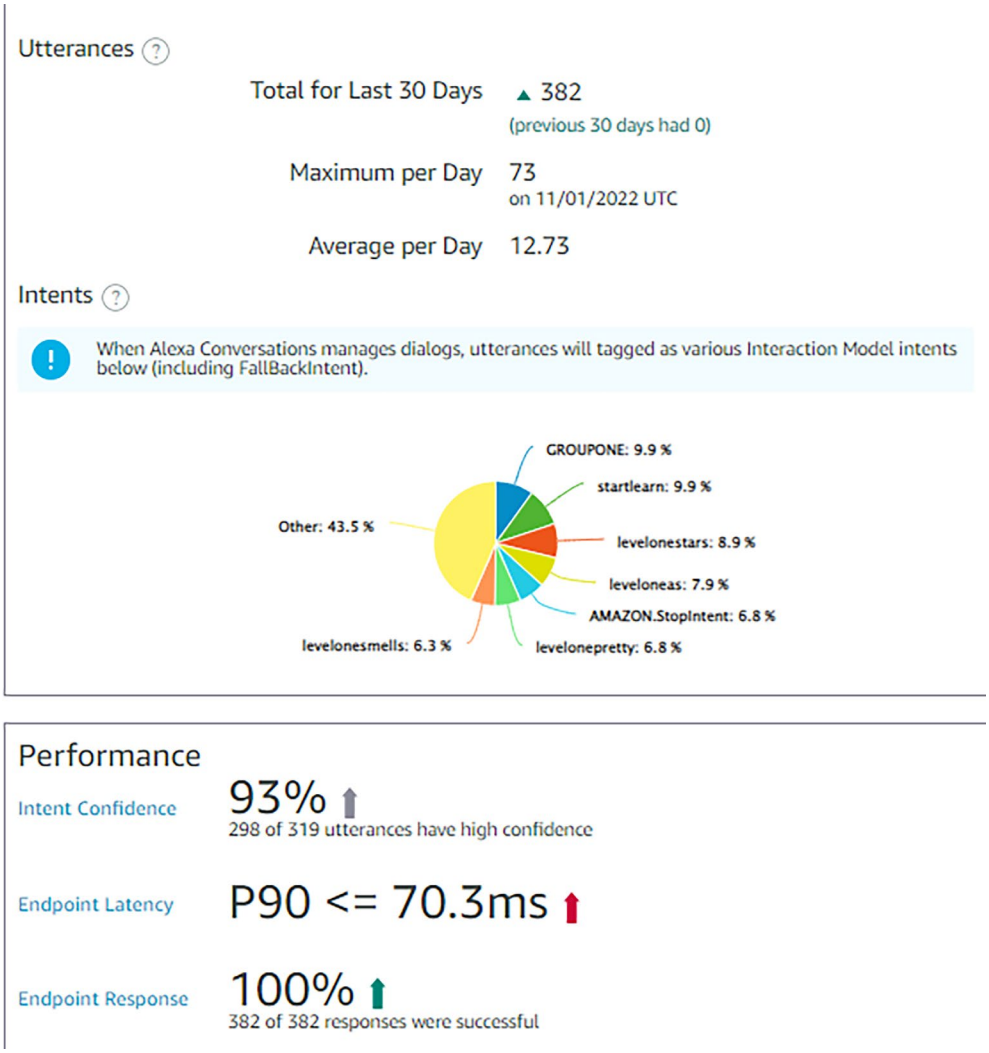



**FIGURE 3** | Students' usage and intent confidence.

considered completed. Reading fluency, accuracy, vocabulary and comprehension were recorded separately.

Accuracy, automaticity, speed, prosody and expression and pronunciation were assessed (reading with accuracy and avoiding making errors or missing words, maintaining a natural speed for reading, pronouncing words correctly, stressing important words, applying intonation and connected-speech rules and using rising and falling intonation patterns to show ideas were linked or coming to an end).

This study was approved by the University Research Ethics Committee, the Ministry of Education, the school and parents and experiment participants.

## 4 | Results

### 4.1 | Homogeneity Between Groups

Prior to conducting the experiment, the homogeneity of the sample was ensured. *t*-tests assessed whether there were differences between the mean scores of students in the experimental and control groups in the pretest of English oral reading fluency skills (Table 2).

The data in each group comprised independent observations and met the assumptions of linearity and normal distribution, confirmed by the Kolmogorov–Smirnov test ($p = 0.182$ for the experimental group; $p = 0.153$ for the control group), with both $p$ values being greater than 0.05. No outliers were detected. The experimental group's skewness was 0.091 and kurtosis 2.405, and the control group's skewness was 0.162 and kurtosis 2.556. Statistical data confirm the two groups were predominantly homogeneous and comparable.

### 4.2 | Oral Reading Fluency Test Results

To answer the research questions, *t*-tests ascertained whether there were differences between the mean scores of students in the experimental and control groups in the post-test of English oral reading fluency skills overall and its subskills. We used $\eta^2$ to measure the impact of using voice chatbot-based AI on English oral reading fluency skills overall and its subskills (Table 3).



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**TABLE 2** | Results of the independent sample *t*-test for ensuring equivalence between experimental and control groups in the pretest of English oral reading fluency skills overall and its subskills.

| Skills | Test | *N* | Mean | SD | *t* | Sig.[a] |
|---|---|---|---|---|---|---|
| Fluency and accuracy | Experimental | 45 | 8.93 | 1.514 | 0.402 | 0.689 |
| | Control | 45 | 8.80 | 1.632 | | |
| Vocabulary | Experimental | 45 | 4.29 | 1.058 | 0.485 | 0.629 |
| | Control | 45 | 4.18 | 1.114 | | |
| Comprehension | Experimental | 45 | 4.02 | 0.812 | −0.129 | 0.898 |
| | Control | 45 | 4.04 | 0.824 | | |
| Overall oral reading fluency skills | Experimental | 45 | 17.24 | 2.814 | 0.374 | 0.709 |
| | Control | 45 | 17.02 | 2.816 | | |

[a]$p > 0.05$ for English oral reading fluency skills overall and its subskills.

**TABLE 3** | Results of the independent sample *t*-test between experimental and control groups in the oral reading fluency skills post-test.

| Skills | Test | *N* | Mean | SD | *t* | Sig.[a] | Effect size $\eta^2$ | Effect |
|---|---|---|---|---|---|---|---|---|
| Fluency and accuracy | Experimental | 45 | 14.82 | 0.747 | 21.800 | 0.000 | 0.915 | Large |
| | Control | 45 | 8.87 | 1.673 | | | | |
| Vocabulary | Experimental | 45 | 8.00 | 0.853 | 17.169 | 0.000 | 0.870 | Large |
| | Control | 45 | 4.69 | 0.973 | | | | |
| Comprehension | Experimental | 45 | 8.51 | 0.869 | 21.021 | 0.000 | 0.909 | Large |
| | Control | 45 | 4.49 | 0.944 | | | | |
| Overall oral reading fluency skills | Experimental | 45 | 31.33 | 1.796 | 27.278 | 0.000 | 0.937 | Large |
| | Control | 45 | 18.04 | 2.730 | | | | |

[a]$p < 0.05$ for English oral reading fluency skills and its subskills.

Figure 4 depicts the differences between the mean scores of the experimental and control groups in the post-test of each subskill and overall English oral reading fluency skills.

#### 4.2.1 | Fluency and Accuracy

The experimental group's mean score was 14.82 with SD of 747, whereas the control group's was a mean of 8.87 with SD of 1.673 in fluency and accuracy post-test. The value of the *t*-test for differences between the experimental and control groups' fluency and accuracy test mean scores was 21.800 with 0.000 significance, which is lower than 0.05, indicating significant differences in favour of the experimental group. To measure the effect of the voice chatbot-based AI on developing fluency and accuracy, $\eta^2$ was used, achieving a high value of 0.915.

#### 4.2.2 | Vocabulary

In the vocabulary post-test, the experimental group's mean score was 8.00 with SD of 0.853, whereas the control group's was a mean of 4.69 with SD of 0.973. The value of the *t*-test for differences between the experimental and control groups' vocabulary test mean scores was 21.800 with a 0.000 significance, which is lower than (0.05), indicating significant differences between groups in favour of the experimental group. We measured the effect of the voice chatbot use on developing vocabulary skills, with $\eta^2$ having achieved a high value of 0.870.

#### 4.2.3 | Comprehension

The experimental group achieved a mean score of 8.51 and SD of 0.869 in the comprehension post-test, whereas the control group's was a mean of 4.49 and SD of 0.944. The value of *t*-test for differences between the experimental and control groups' vocabulary test mean scores was 21.800 with a significance of 0.000 which is lower than (0.05), indicating significant differences between groups in favour of the experimental group. We measured the effect of the voice chatbot on developing comprehension skills using $\eta^2$, which achieved a high value of 0.909.

#### 4.2.4 | Overall Oral Reading Fluency Skills

The experimental group achieved a mean score of 31.33 with SD of 1.796 in the overall oral reading fluency post-test, whereas



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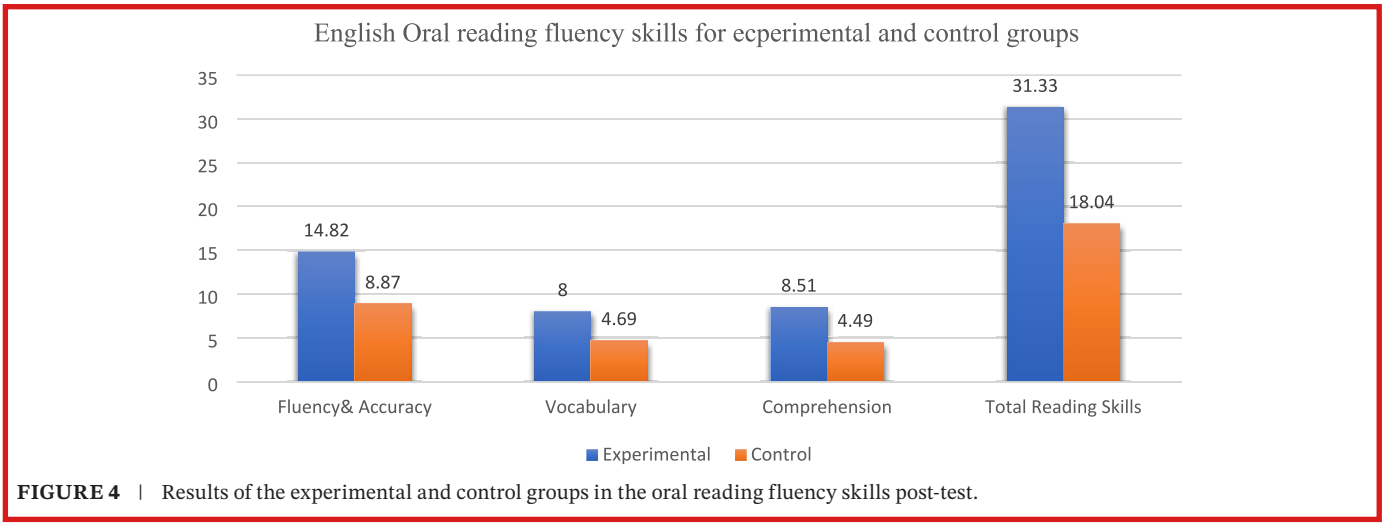



**FIGURE 4** | Results of the experimental and control groups in the oral reading fluency skills post-test.

**TABLE 4** | ANCOVA results between the experimental and control groups.

| Source | SS | df | MS | *F* | *p* |
|---|---|---|---|---|---|
| Corrected model | 3985.069[a] | 2 | 1992.534 | 378.313 | 0.897 |
| Intercept | 1168.519 | 1 | 1168.519 | 221.861 | 0.718 |
| Pretest | 11.691 | 1 | 11.691 | 2.220 | 0.025 |
| Group | 3949.895 | 1 | 3949.895 | 749.947 | 0.896 |
| Error | 458.220 | 87 | 5.267 | | |
| Total | 59,302.000 | 90 | | | |
| Corrected total | 4443.289 | 89 | | | |

[a]$R^2 = 0.897$ (adjusted $R^2 = 0.895$).

the control group's was a mean of 18.04 with SD of 2.730. The value of the *t*-test for differences between the experimental and control groups' overall oral reading fluency skills test mean score was 27.278 with a significance of 0.000 which is lower than (0.05), indicating significant differences between groups in favour of the experimental group which used the voice chatbot-based AI. We measured the effect of the chatbot on developing overall oral reading fluency skills using $\eta^2$, which achieved a high value of 0.937.

Finally, a one-way analysis of covariance (ANCOVA) compared the ratings of the experimental and control groups before and after the post-test. The pretest was used as the covariate, the post-test as the dependent variable and the two groups as the grouping (independent) variables.

Table 4 lists statistically significant differences between the mean scores of the experimental and control groups, with an $F$ value of 749.947 at a significance level of 0.00. The $\eta^2$ value was 0.896, statistically significant at a level of $<0.05$. This indicates a large effect size of voice chatbot-based AI on enhancing oral reading fluency.

Thus, the application helped develop primary school students' fluency, accuracy, vocabulary, comprehension and overall oral reading fluency skills.

### 4.3 | Participants' Interviews

RQ2 addressed participants' perspectives on the use of AI-powered chatbots for learning EFL. Forty students voluntarily participated in semi-structured interviews, which explored participants' perspectives on using such chatbots for oral reading proficiency. The interview data were analysed and interpreted using a deductive qualitative analysis approach (Gilgun 2019), which provided a systematic method for analysing data and answering RQ2. Participants were interviewed in Arabic for accuracy and free expression. The researchers carefully evaluated key points and categorised them into themes. Four categories were extracted from the data through repeated comparisons of students' responses: reducing stress and anxiety, improving engagement and practice, enhancing mastery and efficiency and fun and ease while learning. Table 5 presents thematic codes generated regarding students' perspectives based on the interview data.

The integration of test results into the interview design aligns with a mixed-methods approach, where quantitative performance data inform qualitative exploration. This method ensures a proper understanding of participants' reading abilities, as interviews can reveal contextual factors such as educational background, motivation and reading strategies not evident through tests (Pressley 2000).

Most students reported feeling less anxious when using chatbots:

> When I feared my friends may ridicule me or the teacher might assign me a poor evaluation, I felt constrained. Conversing with the chatbot alleviated my fears, allowing me to express myself more openly than I could with my peers. The chatbot assists in rectifying my English reading skills.
>
> (AR-002)



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**TABLE 5** | Thematic codes generated regarding students' perspectives based on interview data.

| Category | Code | Frequency (%) |
|---|---|---|
| Reducing stress and anxiety | Comfortable and stress-free | 35 (87.5%) |
| | Less nervous and anxious | |
| | Calm while learning | |
| Improving engagement and practice | Feel confident while talking | 33 (82.5%) |
| | Enthusiastic to participate | |
| Enhancing mastery and efficiency | Easily memorize and recall vocabulary | 29 (72.5%) |
| | Fewer linguistic mistakes during oral reading | |
| | Getting immediate support and feedback | |
| Fun and easy while learning | Ease and convenience using chatbots | 31 (77.5%) |
| | Happy, enjoyable, fun dealing with chatbots | |

> I felt less anxious while dealing with a chatbot because it did not compare my English to my classmates'.
>
> (AR-007)

Many participants reported that rehearsing with chatbots instilled confidence to communicate without worry of making mistakes. Correcting mistakes helped them:

> I gained confidence and learned to pronounce a lot of new terms correctly while interacting with the chatbot programme since it provided me with personalized feedback when I made a mistake.
>
> (AR-0018)

> I engaged actively with the chatbot, which motivated me to practice, particularly while I was hesitant to participate in class.
>
> (AR-025)

> Upon making an error during reading, the chatbot provided prompt feedback that facilitated my correction, enhanced my language proficiency, and improved my speaking accuracy.
>
> (AR-027)

Most participants enjoyed interacting with chatbots:

> The chatbot programme reminds me of the mobile games I play in my free time, thus interacting with it brings a smile to my face.
>
> (AR-38)

Participants' feedback indicates that the chatbot program positively impacted their well-being, alleviated their worries and enhanced comfort in mastering vocabulary. Error correction process and rapid feedback facilitated their learning vocabulary, enhanced their reading accuracy and motivated them to practice. This was evidenced by their proficiency and mastery of reading in English.

## 5 | Discussion

This study investigated the impact of an AI-powered chat program on the oral reading fluency of fourth grade primary school children learning EFL, using quantitative and qualitative methodologies to analyse the effects. RQ1 explored the effect of an AI-based chatbot on the accuracy of oral reading in English. Pre- and post-tests were conducted; the experimental group attained better post-test results than the control group, demonstrating considerable improvement in oral reading fluency and reading comprehension. When the impact of the pretest score remained constant, ANCOVA revealed that chatbot learning had a positive impact.

The experimental group's improved oral reading accuracy is ascribable to several factors. The chatbot enhanced students' practice of EFL by using audio clips that could be repeatedly accessed, unlike conventional classroom settings. Continual repetition and rehearing words facilitate mastery of vocabulary, improve verbal interaction and reduce errors. This finding is consistent with studies by Becker and Edalatishams (2019), Gomathy et al. (2021), Koç and Savaş (2024) and Madnani et al. (2019), who indicated that chatbots can enhance both functionality and user experience. Second, as students acclimated to the Alexa chatbot, their apprehension regarding this novel technology was replaced by eagerness to persist with the experience. This aligns with Wu et al.'s (2024) finding that learners need time to adapt to new technology. Congruent with Wu et al.'s (2024) study, students in the experimental group displayed decreased levels of fear and improved confidence in communicating. The prompt feedback offered by Alexa effectively rectified participants' errors, allowing them to articulate freely and amend mistakes without embarrassment. Alexa's feedback mechanism corroborates the conclusions of Guo et al. (2023), Jeon et al. (2023) and



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Kim (2018), who demonstrated that chatbots are effective and convenient instruments for language education.

Alexa's conversational skills and persuasiveness make it an asset in language classrooms (Cook 2021). According to the data, congruent with Dizon's (2020) results, the chatbot's teaching of lower-order skills like letter sound recognition, decoding and phonemic awareness helped students develop higher-order skills like fluency and understanding—the AI system immediately codes and memorizes phrases when displayed on the app screen and detect their significance (Hempenstall 2016). This aligns with findings by Alrajhi (2024), Kim (2018), Kwon et al. (2023), Srinivasan and Murthy (2021) and Yang et al. (2022), who suggested that the accurate assessment provided by AI technology aids in the development of speaking skills when faced with new terms. Studies by Kim and Su (2024), Wu et al. (2024), Yang et al. (2022) and Zou et al. (2020) show that students participated more in English language learning and communication activities with chatbot-based AI technology and their accuracy in pronouncing English words improved.

While the results indicate that the experimental group outperformed the control group on the post-test, they contradict those of Judson (2010) and the Organisation for Economic Co-operation and Development (2015), who found no evidence that technology enhances oral reading fluency and reading comprehension. These anomalous findings can be ascribed to their premature timing as chatbot technologies have not yet reached today's level of sophistication.

RQ2 pertains to the experimental group's perspective on using chatbots for English language acquisition. As with Yang et al.'s (2022) study, the interviews revealed that participants favourably viewed learning via the chatbot. This is consistent with Kwon et al.'s (2023) finding that participants exhibited satisfaction with learning via Google's Dialogflow machine learning. The corroboration can be attributed to the methodology and procedures employed in the current study, which are comparable to those of Kwon et al. (2023) and the substantial similarity in sample size and academic stage.

The chatbot's feedback benefited participants' learning speed and facilitated prompt and effortless error correction. Such software programs foster academic and emotional growth, confidence, comfort and interaction during learning. Most students confirmed that the chatbot facilitated language acquisition, enhanced pronunciation accuracy and mitigated mistake-related anxiety. Students perceived Alexa as a proficient English educator, enabling them to engage without apprehension or limitations, simultaneously receiving assistance with errors and obtaining positive reinforcement. These findings align with studies by Alrajhi (2024), Chien et al. (2022), Kwon et al. (2023), Kim and Su (2024) and Wu et al. (2024), validating the effectiveness of chatbot programs in facilitating acquisition of second-language skills. Students cultivated a habitual and familiar interaction with the Alexa chatbot, which fostered positive emotional connections and enhanced their learning. Learners' perceptions and feelings are indicators of accurate oral reading—Kuhn et al. (2010) emphasized that fluency encompasses several components, including emotions, precision and speed, particularly in connection to prosody.

This study's results have several implications. Alexa, an AI-powered chatbot, helps improve students' oral reading and reading comprehension. There was a favourable association between the level of students' English language mastery and their attitude towards the utilization of the chatbot for language training. Hence, incorporating technology into English language education in Egypt would lead to a considerable reduction in oral reading fluency errors and improvement in reading comprehension. Consistent with Martins and Capellini's (2019) findings, this study's results can be interpreted as an indication that oral reading fluency is an important component of comprehension instruction—vital in helping students understand what they read.

Findings indicate that using automated chat platforms like Alexa enhance oral reading and cultivate favourable dispositions among elementary school students, which underscore the necessity to broaden the application of AI tools, particularly chatbots, among students. As stated by Fryer et al. (2020), the 'golden age' of chatbot learning in L2 learning is 'on the horizon' and active cooperation with digital multimedia learning experts is needed (p. 8). The majority of chatbots function as IPAs, hence tailored programmes for enhancing language proficiency is crucial. The experimental procedures outlined in this paper are important for stakeholders, particularly regarding improving EFL speakers' oral reading.

AI-powered chatbots can be a great asset for EFL students. Expertly crafted software can converse fluently with students, respond to their answers and correct their mistakes in real time while helping them acquire and use the language effectively. This study is, to our knowledge, the first investigation focused on enhancing the accuracy of oral reading in EFL for students whose native language is Arabic.

## 6 | Conclusions and Limitations

While this study provides valuable insights into the effectiveness of AI-powered chatbots in improving oral reading fluency among primary school students, certain limitations should be acknowledged. The sample size was relatively small and limited to a specific educational context, which may impact the generalizability of the findings. Future research should explore larger and more diverse populations to enhance the applicability of these results across different learning environments. The study primarily focused on short-term improvements in reading fluency. The long-term impact of chatbot-assisted learning, including retention of reading skills and continued language development, remains an area for further investigation. Longitudinal studies would be beneficial to assess how AI-driven interventions influence students' reading abilities over time. While the study was conducted within a specific regional context, its findings may have broader implications for other EFL settings. However, cultural and linguistic differences, as well as varying levels of access to technology, may affect the effectiveness of such interventions. Future studies should examine these factors to provide a more comprehensive understanding of AI's role in language education on a global scale.

Subject expectancy, common in experimental research, might have affected the findings. The research instrument's reliability



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may necessitate further validation. However, the limitations did not significantly affect the research outcomes. Future research may necessitate a more comprehensive examination of the impact of chatbots such as Alexa on learning English, including reading, writing and speaking.

Developing an AI-dependent chatbot necessitates an investment of time and effort; Amazon's eventual implementation required a series of stages involving numerous adjustments.

This study's findings align with Egyptian Ministry of Education's initiatives to incorporate AI in English language instruction. The AI chatbot Alexa effectively targeted weaknesses in oral reading and reading comprehension among Egyptian students. Its efficacy was demonstrated by students' improved proficiency in oral reading fluency and reading comprehension.

This study indicates that AI technology in education can enhance students' language proficiency and address deficiencies. Participants' perspectives revealed the extent to which technology augmented their language acquisition, specifically English. It is imperative that the advancement and application of technology be accompanied by a sound command of English and adherence to cultural norms. This study's results, along with the criteria for Alexa, can benefit stakeholders to strategize, construct, implement and assess technological interventions' efficacy in EFL.

Educators must acquire proficiency in using AI to develop instructional materials. This study's methodologies may serve as a resource for instructors to create comparable software applications. It is proposed to conduct training sessions for English language educators to ascertain their perspectives on integrating Alexa into their instructional practices.

This study provides insights into using AI chatbots across various educational levels. While the primary level was this study's focus, exploring the application of chatbots in secondary or higher education could enhance the findings. Using chatbots to generate research or facilitate searches for academic resources across different disciplines, possibly enhancing research capabilities and fostering higher-order cognitive abilities, could also be explored.

### Author Contributions

**Mohamed Ali Nagy Elmaadaway:** conceptualization, methodology, investigation, validation, formal analysis, supervision, resources, project administration, visualization, writing – original draft, writing – review and editing. **Mohamed Elsayed El-Naggar:** conceptualization, methodology, software, investigation, validation, formal analysis, writing – original draft, visualization. **Mohamed Radwan Ibrahim Abouhashesh:** conceptualization, methodology, formal analysis, validation, visualization, resources, writing – review and editing, writing – original draft.

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### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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